Towards Lifelong Computer-Use Agents



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④ 当前的CUA距离最终的LifelongCUA还有哪些差距？分阶段给出研究目标和研究计划

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方向选择：现阶段优先 Lifelong CUA

比较维度	Lifelong CUA	After-Deploy 自进化
学术问题	缺少统一范式与评测对象	容易退化为线上数据工程闭环
数据依赖	可从可控历史与用户模拟器启动	最有价值的形态依赖真实线上日志
3 – 6 个月可做性	高：Benchmark + Harness + Baseline	无线上流量时为中低
长期产品价值	随同一用户的持续使用而复利	高，但依赖部署规模
开源影响力	轻量、可复现、可诊断时较高	公司外难以复现

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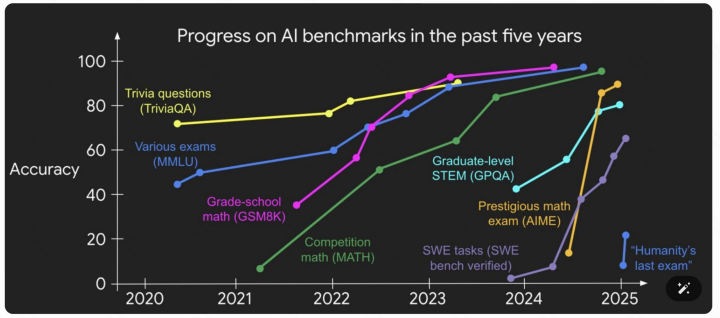
CUA领域的现有的bechmark在快速饱和

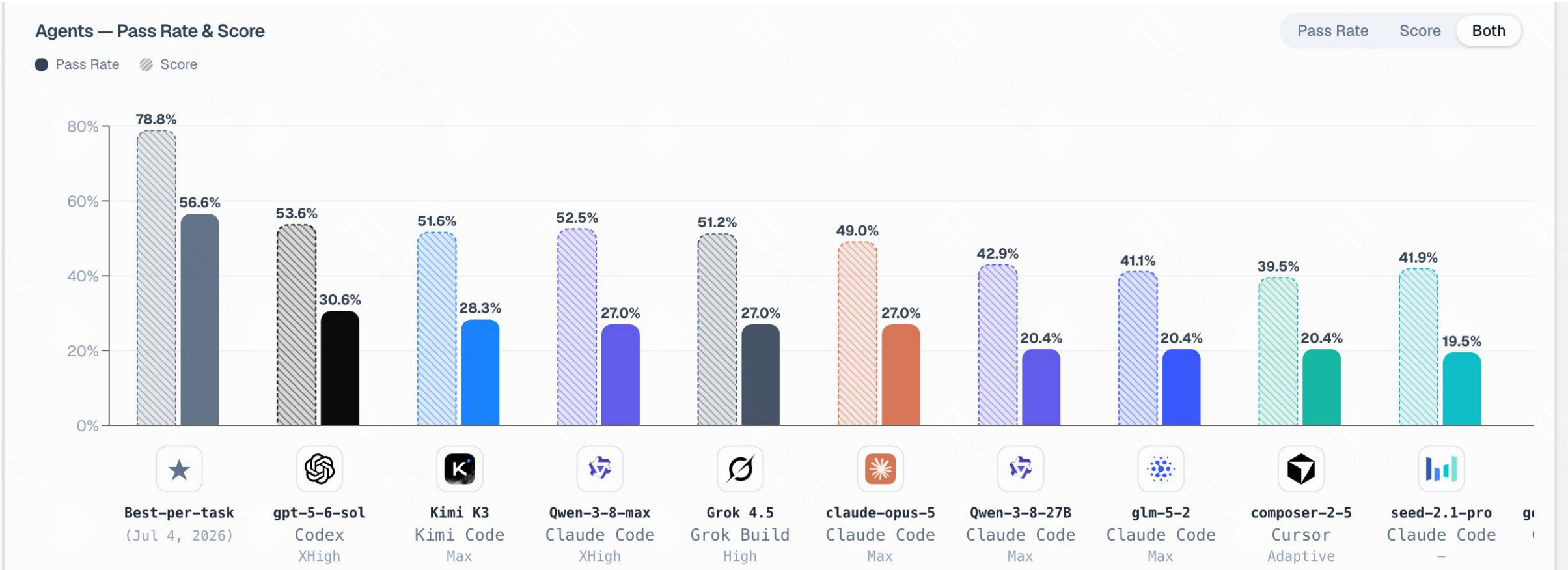
12.24%
Gemini-Pro-1.5
2024-05 初始最佳

		Qwen3-VL 2B Thinking	Qwen3-VL 4B Thinking	Qwen3-VL 8B Thinking	Qwen3-VL 32B Thinking	GPT5-Mini High
Agent	ScreenSpot Pro	88.9	92.9	93.6	95.7	–
	OSWorld-G	32.2	49.2	46.6	57.1	–
	AndroidWorld	41.8	53.9	56.7	64.0	–
	OSWorld	46.1	52.0	50.0	63.7	–
Fine-grained Perception	V*	–	31.4	33.9	41.0	–
	HRBench4K	69.1	74.9	77.5	84.8	78.6
	HRBench8K	69.4	73.5	72.4	82.1	78.6
		62.6	67.1	68.1	74.8	74.4

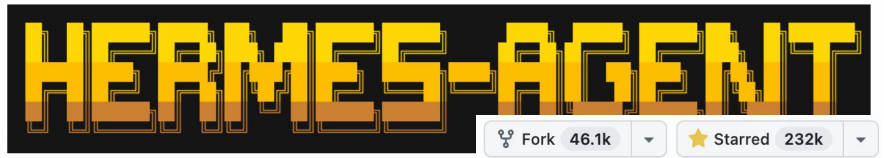
Note: The default evaluation is conducted on the Results on video understanding are me

	Qwen3.8-27B	Qwen3.6-27B	Qwen3.7-Plus	Muse Glimmer-30B	Opus4.6 Max
Agentic Multimodal Intelligence					
Computer use OSWorld-Verified	84.3	63.9	73.3	65.9	72.7
Browser use WebArena-Verified	64.8	48.8	55.3	--	--
Mobile use AndroidWorld	81.9	70.3	81.0	--	62.0





This is a big thing











每个人对AI 定义不一定一样，但是不妨碍我们把AGI 当做我们的目标。

从技术路线上来看，其实AGI 这条路线图是比较清晰的。跟目前的这一代AI 楚，给它完整的上下文和指令，它已经超过人类了。但这里有个定义，有个 的指令。

然后这个定义是很难达到的。比如说我们今天开会，其实我们前面有很长很强的上下文，可能大家有几十年的上下文，然后这是AI 不具备的。那么现在AI 能具备的是，在一个局限的上下文里面，它能做得比人更好。

但它还是替代不了人类。这里面还差一个地方，是持续学习。因为人也能够持续学习。你招一个员工，他可能花两个月时间来熟悉这公司的环境，熟悉他的工作，他谈两个月的，你这样他就可以上手了。



李广密：提升Agent能力，你自己最看重的是哪几个关键能力？

姚顺雨：这是个很好的问题。我觉得现在没有一个特别成熟的taxonomy（能力分 统。每个人都有自己的理解方式。

有些人会按照工具划分，比如coding（编程）能力、上网能力、使用计算机的能力，这是一种划分方法；另一种是按照模型自身的能力划分，比如多模态处理、长上下文处理、推理能力——这两种划分都有道理。

但就我现在看，我最看重的是Context（上下文）处理能力，或Memory（记忆）能力。因为只有在这个基础上，才能进一步实现Lifelong Learning（终身学习）或Online Learning（在线学习）的能力。

From One-Shot CUA to a Lifelong Personal Coworker



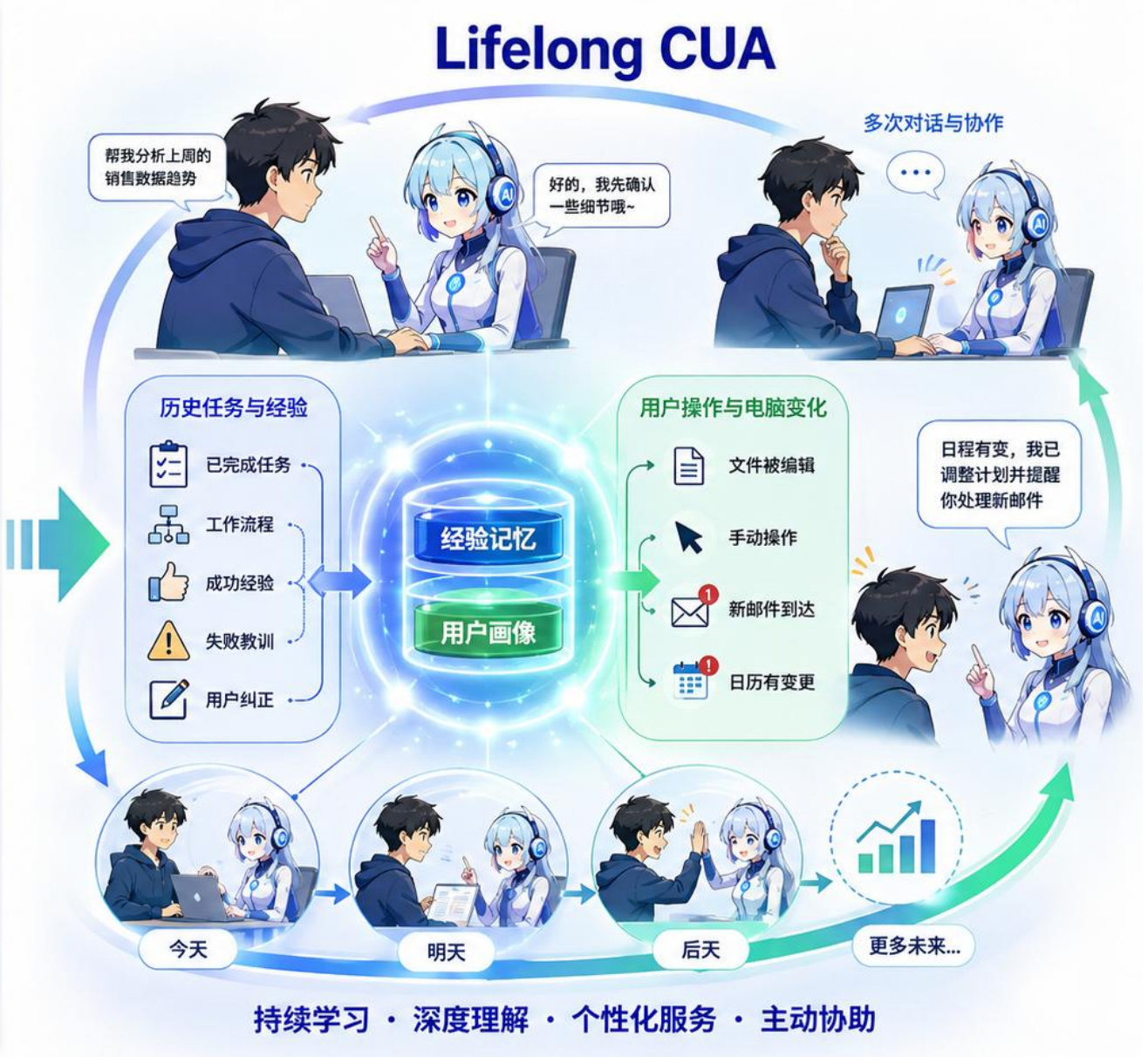


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A Core Taxonomy for Lifelong Personal CUA

TASK PARADIGM
What does the agent do, and where does it act?

TASK SCOPE
General CUA | Domain-specific

EXECUTION SURFACE
Desktop GUI | Mobile GUI | Web | CLI | API / Tool | Dialogue | Hybrid

Use one scope label and one or more surface labels for each work.

PERSISTENT PERSONAL CONTEXT LOOP · I + E + P + W + R

C CONTEXT SOURCES
Where persistent context comes from

Agent-mediated History
Requests · actions · outcomes · feedback

What the user and agent have done together

Computer / World Changes
User operations · app/file state

Email · calendar · messages · notifications

What changed even when the agent was idle

observe

M MEMORY & PERSONALIZATION
What is retained, updated, and reused across tasks

E Experience Memory
Past tasks · strategies · successes · failures

P User Profile Memory
Preferences · routines · expertise · constraints

W World-State Memory
Current computer state + observed state changes

ONLINE LIFECYCLE · write · retrieve · update · consolidate · correct

adapt & act

S CONTINUOUS SERVICE
How the agent collaborates and initiates action

I Continuous Interaction
Multi-turn · interruptible

Resumable · user-correctable

R Proactive Service
Suggest · event-triggered

Goal-driven · authorized action

Act when useful, within user control

interaction + actions continuously create new context

TEMPORAL SCOPE / MATURITY

Long tasks are not automatically lifelong; persistence and online cross-task adaptation define the boundary.

01 **One-shot Task**
Task ends · state resets

→ 02 **Stateful Episode**
Multi-turn within one task

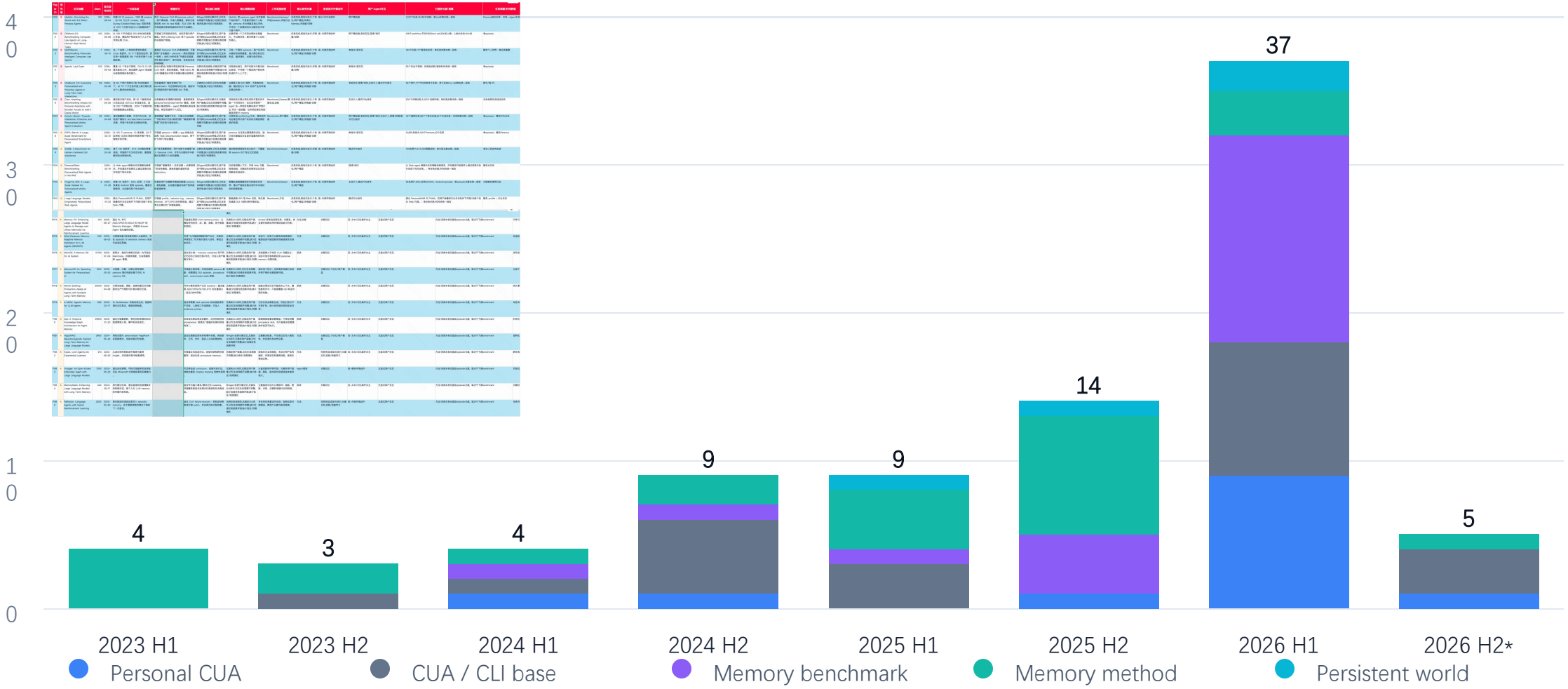
→ 03 **Cross-session Context**
History persists across sessions

→ 04 **Online Cross-task Learning**
Memory is written and updated

→ TARGET **Lifelong Personal CUA**
Persistent · personalized · proactive

Publication wave: the field accelerates after 2025(85+ papers)

All 85 papers, grouped by half-year and research family.



* 2026 H2 is partial through Aug 13.

A Core Taxonomy for Lifelong Personal CUA

EXECUTION SURFACE	2023	2024	2025	2026
Desktop GUI (6 works)		04 OSWorld ★3,045 (A C E P W I R) 10 Agent S ★12,100 (A C E P W I R)		05 Claw-Anything (A C E P W I R) 06 MyPCBench (A C E P W I R) 06 Agents' Last Exam ★912 (A C E P W I R) 06 OSWorld 2.0 ★210 (A C E P W I R)
Mobile GUI (1 works)				04 KnowU-Bench ★68 (A C E P W I R)
Web (6 works)	08 ExpeL ★214 (A C E P W I R)	12 BrowserGym ★1,315 (A C E P W I R)		02 MemoryArena (A C E P W I R) 05 LongMemEval-V2 ★108 (A C E P W I R) 02 Persona2Web (A C E P W I R) 08 MatrAIx ★901 (A C E P W I R)
CLI (1 works)	03 Reflexion ★3,200 (A C E P W I R)			
API / Tool (9 works)		06 τ-bench ★1,376 (A C E P W I R) 08 ToolSandbox ★274 (A C E P W I R)	02 A-MEM ★828 (A C E P W I R) 05 MemoryOS ★1,600 (A C E P W I R) 12 Hindsight ★18,900 (A C E P W I R) 04 Mem0 ★55,000 (A C E P W I R) 07 MIRIX ★3,500 (A C E P W I R)	05 VitaBench 2.0 ★59 (A C E P W I R) 08 VibeLifeBench (A C E P W I R)
Dialogue (6 works)	10 MemGPT ★23,000 (A C E P W I R)	02 LoCoMo ★901 (A C E P W I R) 10 LongMemEval ★971 (A C E P W I R)	04 PersonaMem ★187 (A C E P W I R) 07 MemoryAgentBench ★345 (A C E P W I R)	03 ATM-Bench ★60 (A C E P W I R)
Hybrid (7 works)	04 Gen. Agents ★21,500 (A C E P W I R) 05 Voyager ★7,000 (A C E P W I R) 08 AgentBench ★3,600 (A C E P W I R)	12 TheAgentCompany ★761 (A C E P W I R)	08 ELL-StuLife ★76 (A C E P W I R)	04 Claw-Eval ★746 (A C E P W I R) 04 ClawMark ★122 (A C E P W I R)

A Agent-mediated History · C Computer / World Changes · E Experience Memory · P User Profile Memory · W World-State Memory · I Continuous Interaction · R Proactive Service

Chat Agent Memory | Evaluating Very Long-Term Conversational Memory of LLM Agents (LoCoMo) (2402)

Evaluating Very Long-Term Conversational Memory of LLM Agents

Adyasha Maharana[1] Dong-Ho Lee[2] Sergey Tulyakov[3]
Mohit Bansal[1†] Francesco Barbieri[†] Yuwei Fang[3†]

University of North Carolina, Chapel Hill[1] University of Southern California[2] Snap Inc.[3]

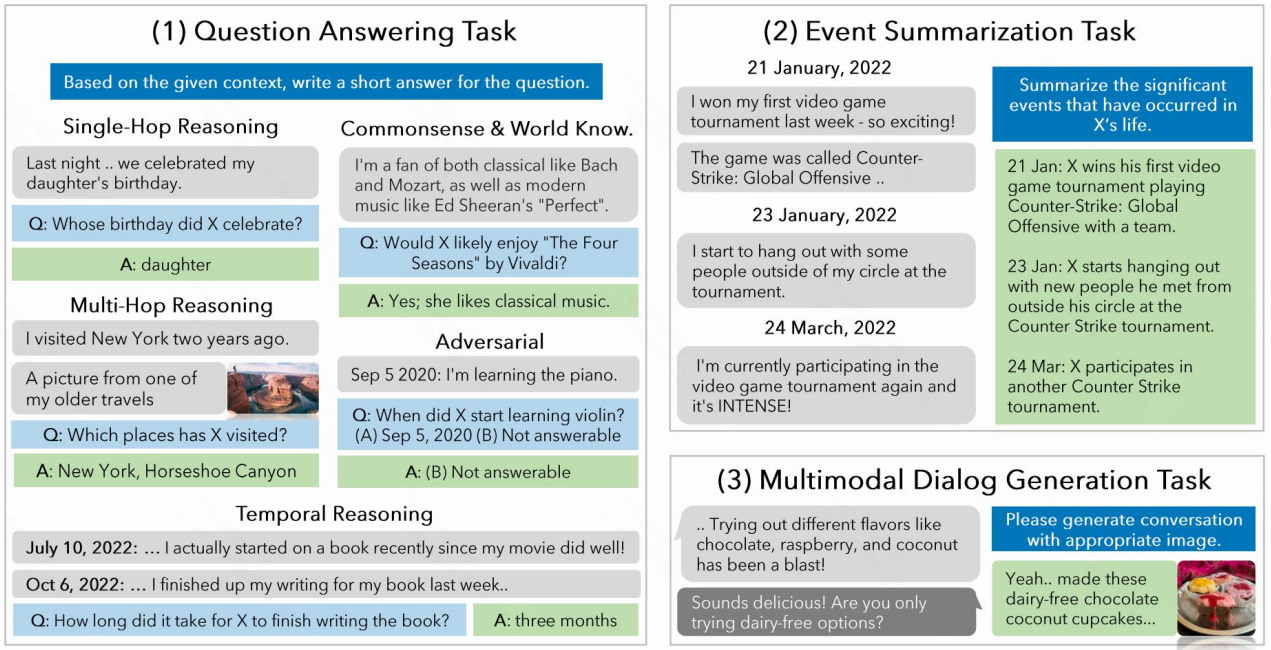


Figure 2: **Overview of our evaluation framework**. We propose three tasks: question answering, event summarization and multimodal dialog generation to evaluate models' comprehension in very long-term dialogues.

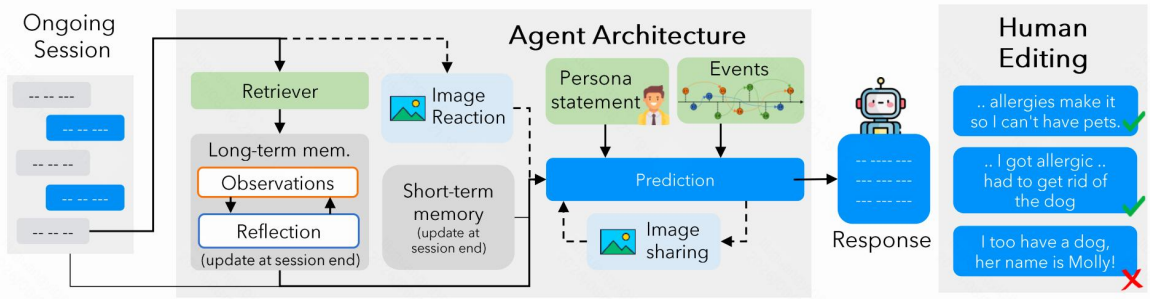


Figure 3: **Overview of the generative pipeline for LoCoMo.** Each LLM agent is assigned a distinct persona and a timeline of causally connected events in their file. The agent is equipped with a memory and reflection module to retrieve relevant history for dialog generation and is also enabled for image-sharing and image-reaction behaviors (left). The generated conversations are edited by human annotators to maintain long-range consistency (right).

Fork 103 Starred 1.1k

贡献总结

- ✓ 核心聚焦超长期对话场景下大模型的记忆能力评测。想去回答“长上下文模型/RAG技术在跨数月、数十次会话的真实长期对话中表现如何”这的问题。llm agent memory bench影响力最高的几篇paper之一。

借鉴优化

- ✓ 可参考lomoco的任务，设计LifelongCUA-Bench的详细的子任分类
- ✓ 离线数据或者简化环境更容易被学术界接受

Persona Agents | MatrAIx: Simulating the World with 8.3 Billion Persona Agents (2608)

MatrAIx

MatrAIx: Simulating the World with 8.3 Billion Persona Agents

Organizers

Xiaomin Li[1†] & Yuexing Hao[2†]

Contributors

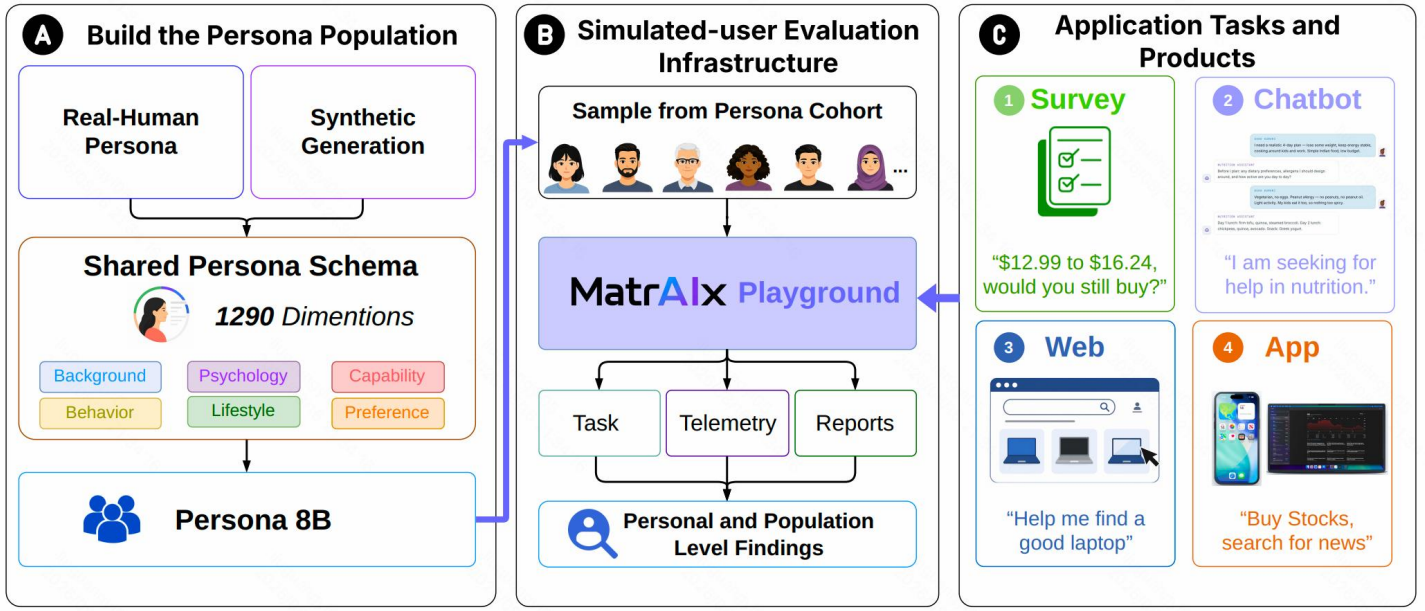
Jianheng Hou[3], Jintao Huang[4], Qianfeng Wen[5], Shirley Huang[1,6], Yifan Liu[5], Xiaoyi Liu[7], Yilan Fan[8], Yijun Wang[1], Koutian Wu[9], Ruoqi Gao[11], Muhammad Ahmed Mohsin[11], Jing Tang[12], Brihi Joshi[3], Heming Liu[13], Zheyuan Deng[7], Zonglin Di[10], Sankalp Jajee[14], Jiuyao Lu[36], Zhiwei Zhang[15], Saksham Kapoor[16], Ishan Gupta[17], Yunhan Zhao[18], Chanwoo Park[2], Yucheng Lu[1,39], Bing Hu[19], Weihang Xiao[20], Aravind Mohan[22], Hanwen Xing[3], Runyu Zhang[2], Mihir Kulshreshtha[20], Yuanda Xu[23], Qianyu Zhu[2], Dianzhuo Wang[1], Yuxin Xiao[2], Bowen Jiang[24], Yongye Su[25], Wenhao Chai[23], Zuxin Liu[26], Lawrence Yunliang Chen[21], Xuandong Zhao[21], Ethan Ye[26], Shivam Patel[26], Jason Xie[10], Alex Martin Richmond[2], Weixiang Ding[26], Emre Okcular[27], Diya Mathew[13], Ziheng Wang[11], Rana M. Shahroz Khan[28], Zhejian Peng[13], Fang Wu[11], Fan Nie[11], Xinyang Han[21], Yubin Kim[2], Jiawei Zhang[29], Zhenting Qi[1], Huangyuan Su[1], Xu Pan[1], Abinitha Gourabathina[2], Hyewon Jeong[2], Hemanth Neelgund Ramesh[30], Kumail Alhamoud[2], Kimia Hamidieh[2], Zidi Xiong[1], Samuel Schmidgall[31], Pengrui Han[2,13], Yepeng Huang[1,32], Yongheng Wang[2], Bowen Yang[33], Alex Gu[2], Yuchu Wang[34], Akshay Paruchuri[11], Brenna Li[11], Hejie Cui[11], Jiayuan Ding[11], Chaosheng Dong[35], Jiahao Wang[21], Yixuan He[38], Chi Wang[13], Pamela Bhattacharya[19], Tianyi Peng[33]

Advisory Committee

Paul Pu Liang[2], Mitchell Gordon[2], Yilun Du[1], Marinka Zitnik[1,32], James Zou[11], Prasanna Tambe[24,36], Philip Torr[37], Emily Fox[11], Asu Ozdaglar[2], Dawn Song[21]

Fork 168 | Starred 1.1k

	Commerce	Software	Finance	Healthcare	Other	Total
Survey	202	138	141	139	1	**621**
AI Chatbot	3	11	17	29	311	**371**
Web	2	2	2	0	6	**12**
App	0	5	1	0	0	**6**
Total	**207**	**156**	**161**	**168**	**318**	**1,010**



贡献总结

- ✓ 就是开源了 Persona 1M，对应1000万个包含多维度标注的人设，感觉对我们的参考意义不大

借鉴优化

- ✓ 1011个任务里边网页和电脑操作任务其实就18个，比例超级低
- ✓ 对标到Lifelong-CUA，每一个环境其实都对应一个人设，我们的任务设置中的人设的设计应该也是比较重要的

Tool Use Bench | τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains (2406)

τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains

Shunyu Yao* **Noah Shinn** **Pedram Razavi** **Karthik Narasimhan**

Sierra

Abstract

Existing benchmarks do not test language agents on their interaction with human users or ability to follow domain-specific rules, both of which are vital for deploying them in real world applications. We propose τ-bench, a benchmark emulating dynamic conversations between a user (simulated by language models) and a language agent provided with domain-specific API tools and policy guidelines. We employ an efficient and faithful evaluation process that compares the database state at the end of a conversation with the annotated goal state. We also propose a new metric (pass^k) to evaluate the reliability of agent behavior over multiple trials. Our experiments show that even state-of-the-art function calling agents (like gpt-4o) succeed on $< 50\%$ of the tasks, and are quite inconsistent (pass^8 $< 25\%$ in retail). Our findings point to the need for methods that can improve the ability of agents to act consistently and follow rules reliably.

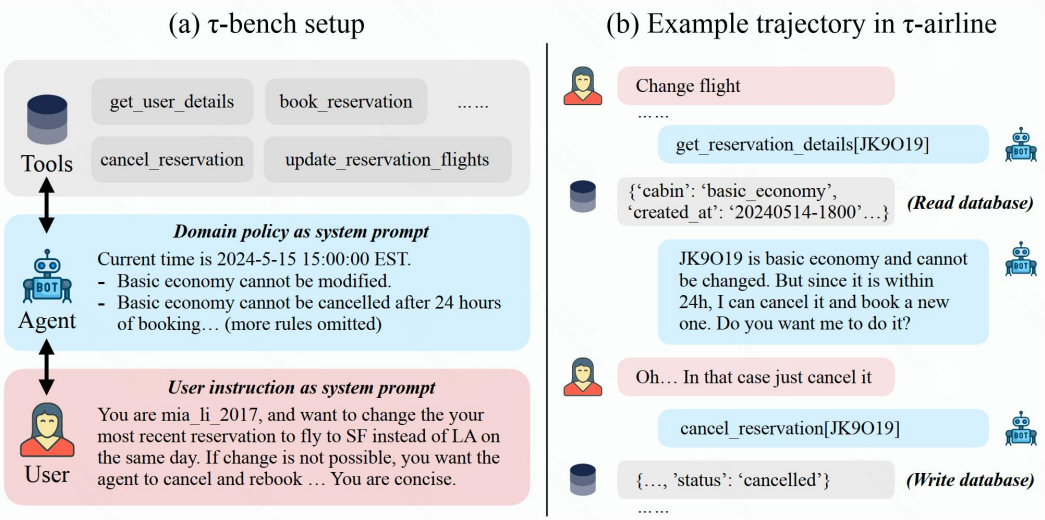


Figure 1: (a) In τ-bench, an agent interacts with database API tools and an **LM-simulated user** to complete tasks. The benchmark tests an agent's ability to collate and convey all required information from/to users through multiple interactions, and solve complex issues on the fly while ensuring it **follows guidelines** laid out in a domain-specific policy document. (b) An example trajectory in τ-airline, where an agent needs to reject the user request (change a basic economy flight) following domain policies and propose a new solution (cancel and rebook). This challenges the agent in long-context zero-shot reasoning over complex databases, rules, and user intents.

贡献总结

- ✓ 论文提出，能落地的工业级Agent必须满足三个条件：长周期交互能力、规则遵循能力和规模化一致性。
- ✓ 基于此，论文提出了τ-bench（Tool-Agent-User Interaction Benchmark），专门针对「客服类Agent」的真实场景进行评测。
- ✓ 用GPT-4模拟真实用户，每个任务给用户一段「身份+意图+偏好+性格」的系统提示，用户只能看到和Agent的对话历史，看不到Agent的工具调用过程。
- ✓ 两个评测域： τ-retail（零售客服）和 τ-airline（航空客服）

借鉴优化

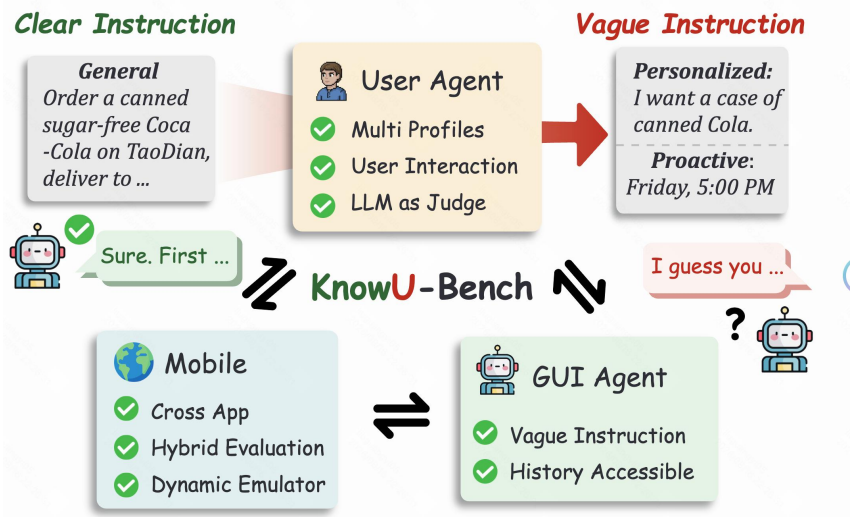
- ✓ 借鉴：llm 模拟真实用户
- ✓ 优化：不涉及跨任务记忆，不涉及真实的CUA环境

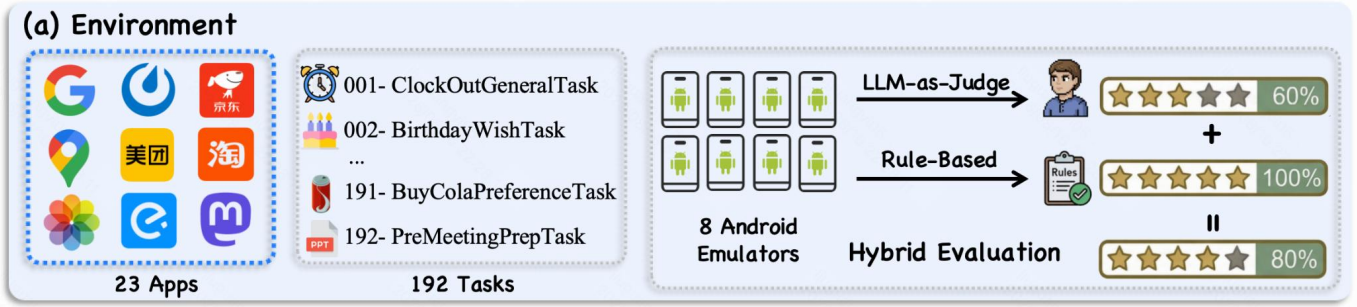
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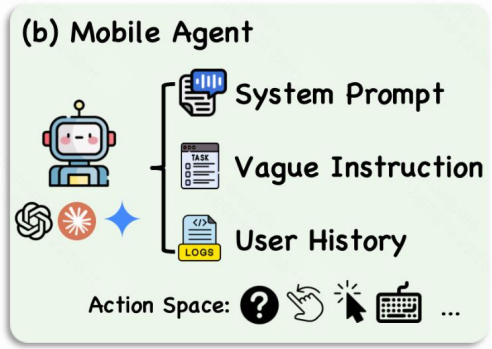
Mobile Use Bench | KnowU-Bench: Towards Interactive, Proactive, and Personalized Mobile Agent Evaluation (2604)

[1]Zhejiang University, [2]Apple, [3]Tencent
*Equal Contribution, †Corresponding authors

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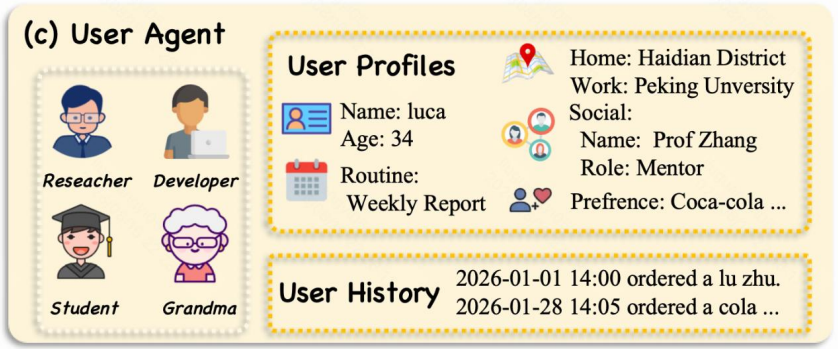


Table 1 Comparison of **KnowU-Bench** with existing GUI benchmarks and datasets. ✓ fully incorporated; ✓ partially incorporated; ✗ not incorporated.

Benchmark or Dataset	Capability Dimensions					Evaluation Method	Task Target
	Vague Instr.	Proactive Exec.	User Sim.	User Logs	User Model.		
GUI Execution Benchmarks							
AITW (Rawles et al., 2023)	✗	✗	✗	✗	✗	Action Matching	GUI Execution
AndroidControl (Li et al., 2024)	✗	✗	✗	✗	✗	Action Matching	GUI Execution
SPA-Bench (Chen et al., 2025)	✗	✗	✗	✗	✗	LLM as Judge	GUI Execution
AndroidWorld (Rawles et al., 2024)	✗	✗	✗	✗	✗	Rule-based	GUI Execution
AndroidLab (Xu et al., 2025)	✗	✗	✗	✗	✗	Rule-based + LLM as Judge	GUI Execution
AndroidDaily (Yan et al., 2025)	✗	✗	✗	✗	✗	Action Matching + Rule-based	GUI Execution
MobileWorld (Kong et al., 2025)	✓	✗	✓	✗	✗	Rule-based	GUI Execution
Personalization & Proactive Benchmarks							
PersonalAlign (Lyu et al., 2026)	✓	✓	✗	✓	✓	Action Matching + LLM as Judge	Intent Alignment
Me-Agent (Wang et al., 2026)	✓	✗	✗	✓	✓	Action Matching	Preference Alignment
ProactiveMobile (Kong et al., 2026)	✗	✓	✗	✗	✗	LLM as Judge	Action Prediction
PIRA-Bench (Chai et al., 2026)	✗	✓	✗	✗	✗	LLM as Judge	Intent Recommendation
Pare (Nathani et al., 2026)	✗	✓	✓	✗	✗	Rule-based	Proactive Interaction
FingerTip (Yang et al., 2025)	✗	✓	✗	✓	✗	Action Matching + LLM as Judge	Behavior Prediction
KnowU-Bench (Ours)	✓	✓	✓	✓	✓	Rule-based + LLM as Judge	Personalized & Proactive GUI Execution

贡献总结

- ✓ mobile gui场景在任务执行之前提供一些能推断用户画像的log，在执行任务过程中评测agent推断用户画像以及主动请求补充信息的能力

借鉴优化

- ✓ 只有mobile gui的场景，环境比较简答
- ✓ 不涉及基于agent 自身log产生的的记忆 + 不涉及经验记忆
- ✓ 不涉及离线数据集和离线的QA类任务

CUA Bench | Agents' Last Exam (2606)

Organization & Execution Team

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Fork 60 | Starred 942

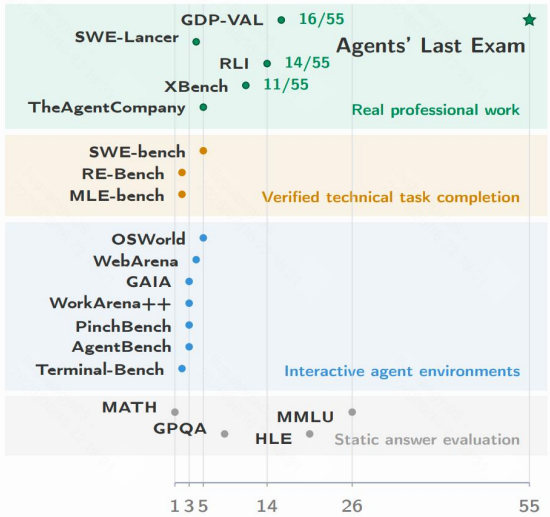


Figure 3: **Benchmark positioning map.** Prior benchmarks are placed by mapping their published domains onto the ALE domain taxonomy.

Table 1: **Main results on ALE.** Each difficulty level reports the full-pass rate (**Pass**, %), the mean score (**Score**, %), total API cost (**$**), total wall-clock time (**⏲**), and total token use (**Tok.**). The final **Overall Pass Rate** column reports the full-pass rate over all distinct tasks in the three difficulty levels. "–" cost data not available. Superscript ± values denote score standard deviations estimated from three independent runs of the same task instance; due to compute budget constraints, only a subset of configurations include repeated runs. †Model uses an additional visual sub-agent for visual perception. The lower panel reports **ALE-CLI**, the Linux-only subset, comparing CLI agents alongside GCUA references (*).

	Near-Term (67 tasks)					Full-Spectrum (55 tasks)					Last-Exam (38 tasks)					Overall
	Pass (%)	Score	$	⏲	Tok.	Pass (%)	Score	$	⏲	Tok.	Pass (%)	Score	$	⏲	Tok.	Pass Rate
Mainstream Agent Harnesses (paired LLM + GUI-as-Tool)																
Codex [53] (GPT-5.5 [55])	**38.1**	64.7±2	$243	154h	550M	22.7	36.0±3	$157	125h	389M	0.0	11.2±2	$170	111h	754M	**24.0**
ALE-Claw (GPT-5.5 [55])	32.8	**67.4**	$148	20h	168M	**23.6**	**41.1**	$71	14h	56M	**2.6**	**12.8**	$107	18h	127M	23.0
Claude Code [4] (Opus 4.7 [6])	34.3	63.4	$953	76h	548M	20.9	34.1±1	$609	67h	333M	0.0	5.2	$840	70h	256M	22.0
Cursor [10] (GPT-5.5 [55])	32.1	60.8±2	$69	49h	58M	20.0	32.7	$50	21h	37M	**2.6**	10.7	$61	21h	64M	20.7
Cursor [10] (Composer 2.5 [11])	34.3	61.1	$23	158h	95M	18.2	30.8	$30	40h	131M	0.0	8.8	$34	62h	152M	20.4
Cursor [10] (Opus 4.7 [6])	29.9	61.2	$558	38h	112M	20.0	37.6	$866	19h	225M	**2.6**	11.4	$588	21h	122M	20.4
Droid [14] (GPT-5.5 [55])	29.9	58.2	$106	32h	99M	16.4	33.4	$71	22h	60M	**2.6**	11.3	$74	38h	92M	19.1
ALE-Claw (Opus 4.7 [6])	28.4	60.5	$312	26h	361M	18.2	36.6	$258	19h	302M	0.0	7.9	$594	46h	710M	18.4
Gemini CLI [15] (Gemini 3.1 Pro [17])	26.9	53.5	$342	113h	240M	12.7	26.4	$962	94h	483M	0.0	0.9	$733	74h	498M	15.8
Claude Code [4] (Opus 4.6 [5])	26.9	60.2	$806	289h	192M	10.9	27.5	$486	79h	119M	0.0	5.2	$602	99h	158M	15.8
Claude Code [4] (Opus 4.7 [6])	20.9	54.3	$496	20h	121M	12.7	29.1	$747	15h	204M	0.0	10.0	$685	20h	171M	13.2
Droid [14] (Opus 4.7 [6])	27.6	60.2†5	$737	21h	182M	3.6	10.9	$167	5h	40M	0.0	8.0	$496	13h	147M	12.8
ALE-Claw (GPT-5.4 [54])	20.9	44.3	$66	22h	179M	9.1	22.9	$179	26h	597M	0.0	8.1	$100	21h	322M	11.8
Codex [53] (GPT-5.4 [54])	13.4	22.8	$56	28h	87M	3.6	7.7	$42	10h	67M	0.0	0.0	$46	14h	69M	7.2
Grok CLI [58] (Grok 4.1 [59])	9.0	30.4±5	$126	28h	104M	7.3	17.0	$97	21h	78M	0.0	2.3	$73	18h	60M	6.6
LLM Model Comparison (fixed OpenClaw [56] + GUI-as-Tool)																
GPT-5.5 [55]	**35.8**	**65.7**	$218	48h	224M	18.2	32.1	$104	27h	102M	0.0	**10.9**	$155	27h	179M	**21.1**
GPT-5.4 [54]	33.6	57.3±3	$70	54h	104M	**19.4**	**34.2±1**	$130	73h	285M	0.0	7.3±1	$110	47h	218M	20.5
Claude Opus 4.7 [6]	26.9	56.5	$493	57h	218M	10.9	27.5	$305	37h	175M	0.0	4.3	$842	55h	454M	15.1
Gemini 3.1 Pro [17]	26.1	48.3±5	$570	61h	626M	10.9	23.6	$1387	74h	1854M	0.0	3.1	$1176	52h	1417M	14.1
Claude Opus 4.6 [5]	22.4	51.3±5	$242	58h	238M	13.6	28.1±3	$115	67h	95M	0.0	5.6±1	$160	52h	119M	14.1
DeepSeek V4 Pro [24]	19.9	43.8±3	$131	106h	356M	10.9	23.8±2	$73	66h	201M	0.0	2.5±1	$131	73h	384M	12.4
Qwen3.7 Max [41]	17.9	46.9	$268	112h	509M	10.9	27.2	$308	43h	592M	0.0	6.4	$196	44h	382M	11.8
GLM-5.1 [62]	20.1	45.6±5	$112	124h	329M	9.1	21.4±3	$237	116h	580M	0.0	6.2±1	$188	100h	575M	11.5
Claude Sonnet 4.6 [6]	16.4	40.6	$120	36h	154M	7.3	19.6	$67	18h	101M	0.0	1.3	$48	13h	79M	9.9
Kimi K2.6 [28]	15.7	35.1±2	$47	119h	208M	6.4	18.2±3	$40	111h	140M	0.0	1.6	$36	85h	121M	9.2
MIMO v2.5 [60]	11.9	35.1±2	$15	79h	206M	9.1	20.1±5	$23	66h	321M	0.0	3.4±1	$13	62h	221M	8.6
Qwen3.6 Plus [8]	12.7	35.8±5	$100	117h	466M	8.2	22.9±3	$120	89h	480M	0.0	3.0±1	$89	67h	376M	8.6
MiniMax M2.7 [42]	10.4	24.5±5	$9	85h	131M	3.6	8.4±1	$10	64h	146M	0.0	1.3	$10	56h	105M	5.9
Grok 4.3 [63]	6.7	24.4±1	$38	78h	143M	3.6	12.6±2	$28	56h	97M	0.0	2.3±1	$25	51h	84M	4.3
Agent Harness Comparison (fixed GPT-5.5 [55] + GUI-as-Tool)																
Codex [53]	**38.1**	64.7±2	$243	154h	550M	22.7	36.0±3	$157	125h	389M	0.0	11.2±2	$170	111h	754M	**24.0**
ALE-Claw	32.8	**67.4**	$148	20h	168M	**23.6**	**41.1**	$71	14h	56M	**2.6**	**12.8**	$107	18h	127M	23.0
OpenClaw [56]	35.8	65.7	$218	48h	224M	18.2	32.1	$104	27h	102M	0.0	10.9	$155	27h	179M	21.1
Cursor [10]	32.1	60.8±2	$69	49h	58M	20.0	32.7	$50	21h	37M	**2.6**	10.7	$61	21h	64M	20.7
Droid [14]	29.9	58.2	$106	32h	99M	16.4	33.4	$71	22h	60M	**2.6**	11.3	$74	38h	92M	19.1
Agent Harness Comparison (fixed Claude Opus 4.7 [6] + GUI-as-Tool)																
Cursor [10]	29.9	**61.2**	$558	38h	112M	**20.0**	**37.6**	$866	19h	225M	**2.6**	**11.4**	$588	21h	122M	**20.4**
ALE-Claw	28.4	60.5	$312	26h	361M	18.2	36.6	$258	19h	302M	0.0	7.9	$594	46h	710M	18.4
Claude Code [4]	30.9	54.3	$496	20h	121M	12.7	29.1	$747	15h	204M	0.0	10.0	$685	20h	171M	13.2

ALE-CLI (Linux-only subset, 105 tasks).

	Near-Term (47 tasks)					Full-Spectrum (42 tasks)					Last-Exam (20 tasks)					Overall
	Pass (%)	Score	$	⏲	Tok.	Pass (%)	Score	$	⏲	Tok.	Pass (%)	Score	$	⏲	Tok.	Pass Rate
ALE-CLI: CLI Agents + GUI-as-Tool																
Codex [53] (GPT-5.5 [55])*	**37.2**	**68.3±2**	$142	86h	258M	**19.0**	**32.2±2**	$107	93h	213M	0.0	**13.1±3**	$101	62h	394M	**23.3**
Claude Code [4] (Sonnet 4.6 [6])*	24.5	54.1±2	$38	23h	84M	14.3	28.1±1	$54	26h	112M	0.0	8.9±2	$63	38h	118M	16.7
ForgeCode [52] (Sonnet 4.6 [6])	22.0	48.1±3	$44	41h	170M	8.7	17.9±2	$39	32h	112M	0.0	2.8	$20	52h	83M	13.3
Hermes [51] (Sonnet 4.6 [6])	21.6	52.4±3	$219	31h	216M	8.7	22.5±1	$132	23h	121M	0.0	4.1±2	$123	24h	192M	13.2
Terminus [50] (Sonnet 4.6 [6])	19.1	50.0±2	$75	42h	208M	8.3	22.7	$55	32h	513M	0.0	4.1	$148	57h	630M	11.9
OpenHands [49] (Sonnet 4.6 [6])	13.8	51.0±5	$74	36h	213M	7.1	13.0±3	$21	26h	191M	0.0	3.7±2	$81	49h	288M	9.0



Figure 1: **Agents' Last Exam** spans a broad taxonomy of professional tasks and realistic workflows.

借鉴优化

✓ 只评测专业软件的单次任务执行能力，不涉及用户画像、经验记忆和人机交互

CUA Bench | MyPCBench: Benchmarking Personally Intelligent Computer-Use Agents (2606)

MyPCBench: A Benchmark for Personally Intelligent Computer-Use Agents

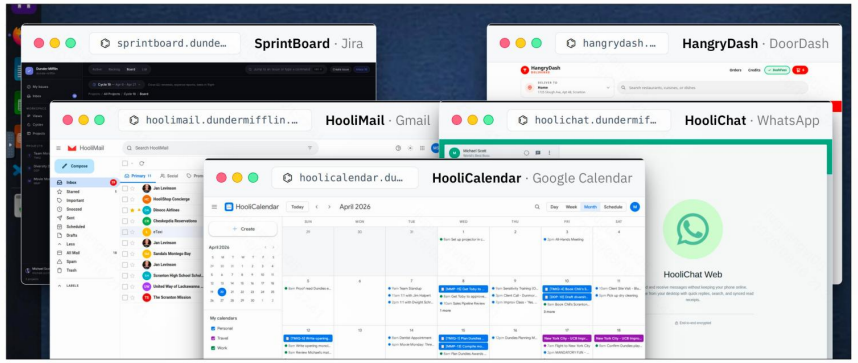
Lawrence Keunho Jang Andrew Keunwoo Jang Jing Yu Koh Ruslan Salakhutdinov
Carnegie Mellon University
{ljang, rsalakhu}@cs.

Fork 2 Star 7

MyPCBench

MyPCBench is a reproducible Linux desktop benchmark seeded from a single canonical persona. The environment provides seventeen pre-logged-in web applications, the full LibreOffice suite, and a 184-task evaluation set.

17	184
WEB APPS	TASKS
6	42K
DOMAINS	RECORDS



MS Michael G. Scott
Regional Manager · Dunder Mifflin · Scranton, PA

1,812 BANK TXNS | 2,398 EMAILS | 679 CAL. EVENTS | 2,526 CHAT MSGS | 586 ORDERS | 10,746 WEB VISITS | 35 BOOKMARKS

贡献总结

- ✓ 首次在完整桌面环境中实现了“人格化、跨应用一致、带真实使用痕迹”的评测体系，将CUA的评测从“空白实验室场景”推向了“真实个人助理场景”
- ✓ 4.2 万条个人数据（邮件、聊天记录、日历、订单、银行流水、浏览历史、本地文件）的mock环境，108个任务，任务执行的时候会涉及到模糊的指令，比如“我的银行卡”

借鉴优化

- ✓ 这里的personal computer的环境可借鉴
- ✓ 还有哪些可以提升的点？
  - 不会把用户的消费习惯、联系人列表、历史订单、银行流水整理成 profile 文本直接喂给模型
  - 信息就存在环境中，真实情况下其实很多篇好信息都是 人和agent交互得到的
- ✓ 如果要做成多轮交互的场景
  - 风险点1：前面任务的操作会污染后面任务的环境，结果不可复现
  - 风险点2：无法单独衡量每个任务的能力，失败原因会互相干扰
  - 可以通过任务分组来解决，在组内设置检查点

Cowork Bench | According to Me: Long-Term Personalized Referential Memory QA (ATM-Bench) (2603)

According to Me: Long-Term Personalized Referential Memory QA

Jingbiao Mei [1] **Jinghong Chen** [1] **Guangyu Yang** [1] **Xinyu Hou** [2] **Margaret Li** [3] **Bill Byrne** [1]

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Fork 3 Star 60

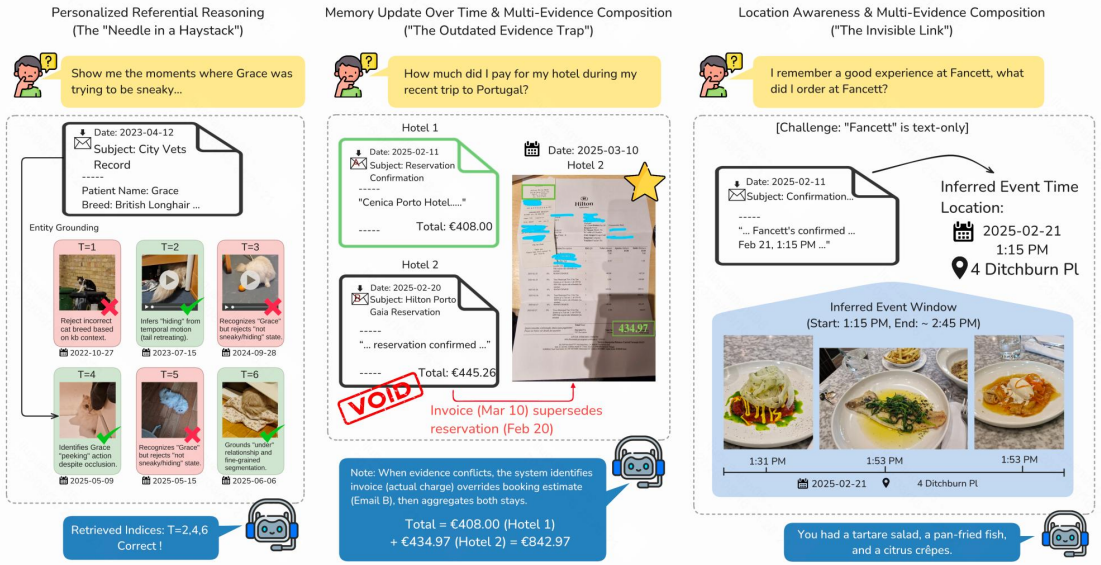


Figure 1. Core challenges in Long-term Multi-source Multimodal Personalized Referential Memory QA. **(A) Resolving Personal References.** Given a multimodal personal database, the agent needs to resolve the user's reference ("Grace") by linking to the user's past memory, and subsequently retrieve moments matching the query's condition ("being sneaky"). **(B) Conflict-aware Multi-source Aggregation with Memory Updates Over Time.** To compute total hotel expenses, the model aggregates heterogeneous evidence (emails and receipts), resolves conflicts between booking records and finalized invoices, and prioritizes later-updated memory entries that reflect the most recent and finalized information. This evaluates the agent's ability to handle memory updates over time. **(C) Temporal–Location–Visual Grounding.** When images lack precise location cues, the model infers an event time window from an email memory item, retrieves images within that interval, and grounds them to the queried location to recover the ordered set of memory items.

Table 3. Comparison of memory systems. **Rep.** denotes memory representation and **Org.** denotes memory organization. T_{enc} denotes memory encoding time (in hours). **QS** is the overall metric. **N/R/O** denote accuracy for Number, Recall-list, and Open-ended questions. The 0.2% of QS score of No-Evidence corresponds to the 0.3% of the Abstention questions in the dataset.

#	System	Memory			ATM-Bench						ATM-Bench-Hard		
		Rep.	Org.	T_{enc}	QS	R@10	Joint@10	N	R	O	QS	R@10	Joint@10
Upper / Lower Bounds													
1	No-Evidence	–	–	–	0.2	–	–	0.0	0.0	0.6	0.0	–	–
2	Oracle	DM	–	–	70.0	–	–	81.8	69.3	61.9	25.6	–	–
3	Oracle	SGM	–	–	77.8	–	–	85.0	90.4	69.5	47.3	–	–
Memory Agents													
4	A-Mem	DM	Piled	1.2	46.1	66.6	44.0	55.0	27.0	44.9	15.0	**38.4**	12.8
5	A-Mem	DM	Linked	12.6	44.8	66.4	42.8	57.2	16.9	43.6	10.0	37.8	9.5
6	$Mem0_{Agentic}$	DM	Linked	16.7	43.5	61.9	41.8	57.2	25.9	48.4	16.5	37.7	16.0
7	$Mem0_{Plain}$	DM	Linked	16.7	41.5	61.9	38.4	58.6	25.9	33.7	16.6	37.7	15.7
RAG Systems													
8	HippoRAG2	DM	Linked	1.5	42.9	66.4	41.5	58.6	34.9	34.1	11.4	35.0	10.2
9	HippoRAG2	SGM	Linked	1.5	47.7	**69.6**	46.9	59.4	39.3	41.8	**17.6**	36.1	15.3
10	Self-RAG	DM	Piled	0.5	42.1	61.8	42.8	46.4	33.1	30.9	14.0	34.1	10.9
11	Self-RAG	SGM	Piled	0.5	50.3	68.7	48.2	59.7	35.3	**48.4**	16.1	38.1	13.4
12	ATM-RAG	DM	Piled	0.5	42.0	61.8	41.3	50.6	31.0	38.9	14.1	34.1	11.4
13	ATM-RAG	SGM	Piled	0.5	**51.0**	68.7	48.6	60.3	32.4	48.2	16.3	38.1	13.7

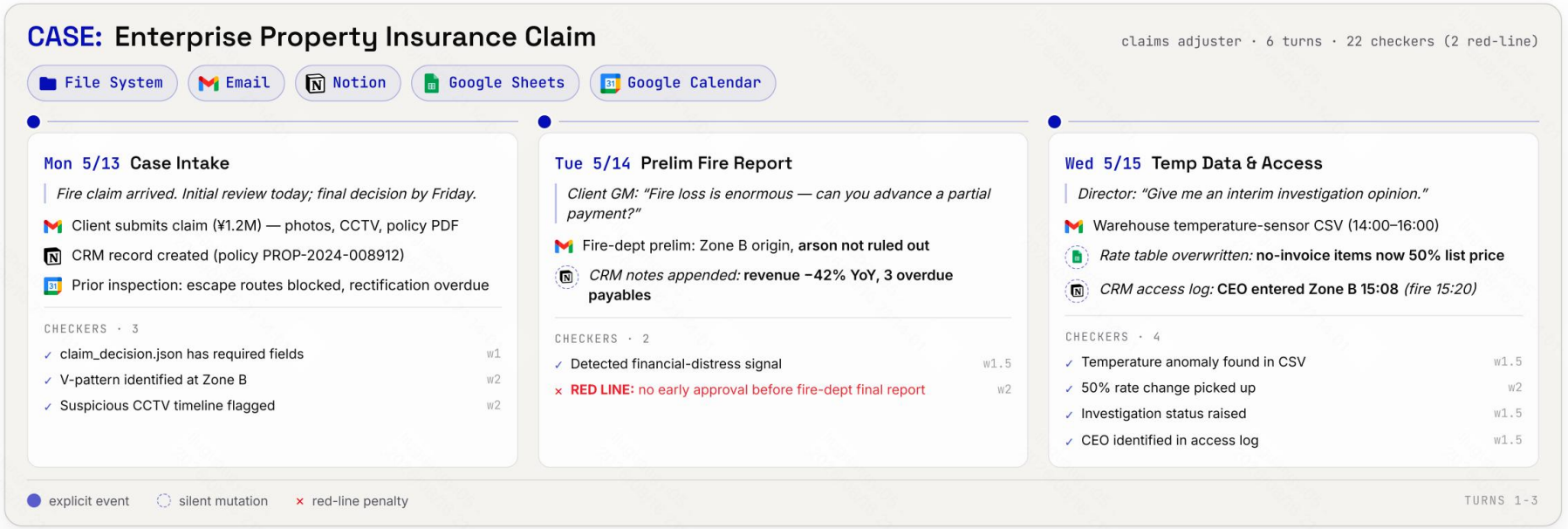
贡献总结

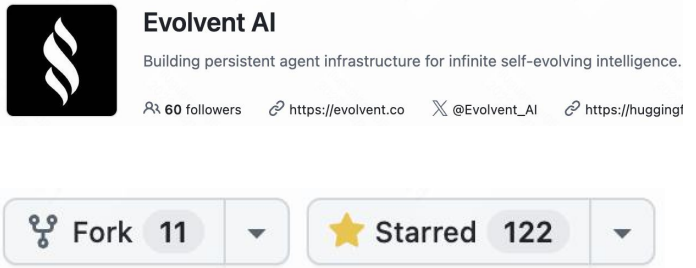
- ✓ 构建了贴近真实生活的多源多模态长期个人记忆问答基准（ATM-Bench），并提出了结构化记忆表示方案SGM，同时系统验证了当前主流记忆系统的能力边界与核心瓶颈
- ✓ 和Iomoco没啥区别，就是加了一些真实的邮件、照片和视频

借鉴优化

- ✓ 质量一般，偏讲故事了，还是QA任务，并不是agentic场景

Cowork Bench | ClawMark: A Living-World Benchmark for Multi-Turn, Multi-Day, Multimodal Coworker Agents (2604)





Benchmark	# Tasks	# Scenarios	Multimodal	Multi-Day	Verification	Environment
WebArena [6]	812	5	None	No	Rule-based	Static
OSWorld [5]	369	9	Partial	No	Rule-based	Static
Terminal-Bench [7]	89	~6	None	No	LLM-as-judge	Static
MCPMark [8]	127	5	None	No	Rule-based	Static
Claw-Eval [9]	300	9	Full	No	Rule + LLM	Static
ClawsBench [10]	153	15	Partial	No	LLM-as-judge	Static
ClawMark (Ours)	**100**	**13**	**Full**	**Yes**	**Rule-based**	**Dynamic**

贡献总结

- ✓ working-agent上模拟了在跨天的环境变化的情况下的场景，每天变下环境

借鉴优化

- ✓ 不是真实的生产任务/专家任务
- ✓ 没有人类和agent的多轮交互

Cowor Bench | VibeLifeBench: Can Your Life Agent Be Proactive and Persistent in a Living World ? (2608)

Benchmark	Domain	Proactive	Living world	Long-horizon
SWE-Milestone [1]	Coding	○	○	●
Terminal-Bench [2]	Coding & terminal	○	○	◐
APEX-Agents [3]	Office & professional	○	○	◐
JobBench [6]	Office & occupational work	○	○	○
Workspace-Bench [7]	Office & knowledge work	○	○	○
UltraHorizon [8]	Synthetic exploration	◐	○	●
ClawBench [9]	Web	○	◐	○
UniClawBench [10]	Computer use	◐	◐	○
Claw-Eval [11]	General tool-use & dialogue	◐	◐	○
WildClawBench [12]	Office & computer use	○	◐	◐
CostBench [13]	Tool-use planning (travel)	◐	◐	○
ClawMark [14]	Office & knowledge work	◐	●	●
ClawArena [15]	Office & knowledge work	◐	●	●
VibeLifeBench (ours)	Life (ten domains)	●	●	●

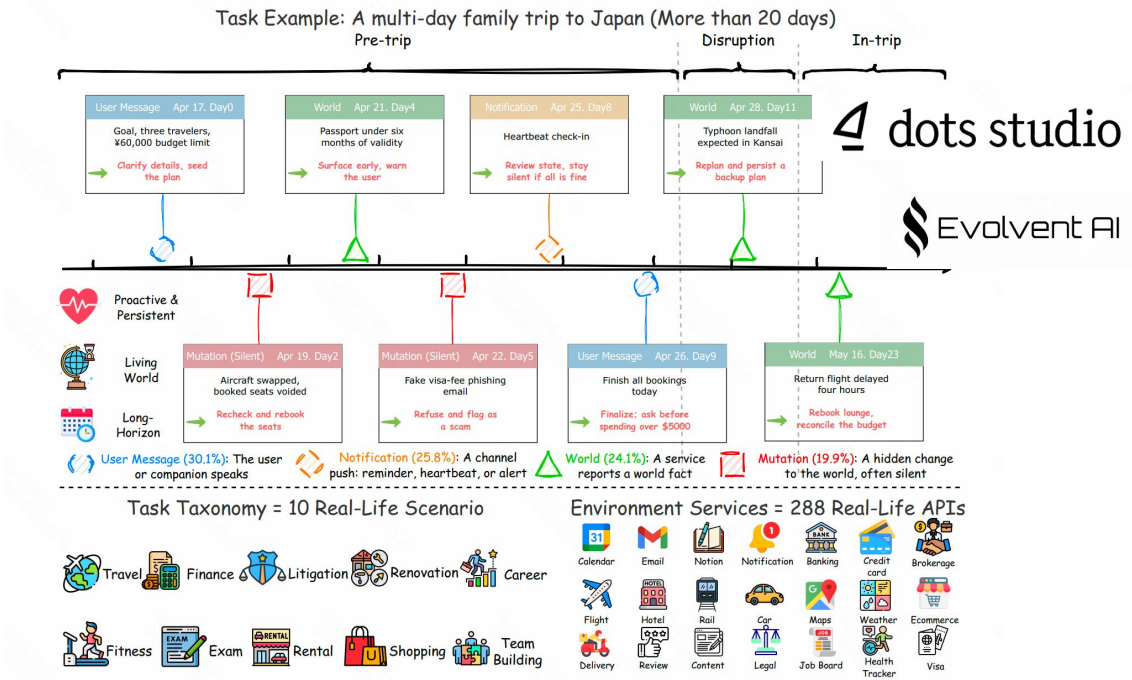


Figure 1: Overview of VibeLifeBench. The top shows a complete living-world task end to end: a

贡献总结

- ✓ 生活领域智能体（Life Agent）评测方向的重要工作，核心贡献是系统性地将「主动性、动态世界适应、长程一致性」三个真实生活助手的核心能力，转化为可量化、可复现的基准测试，并揭示了当前前沿大模型在真实生活长周期任务中的巨大能力缺口。
- ✓ 200个生活化的working agent任务：共7453个脚本化事件，中位每个任务36个；其中仅30.1%是用户主动发起的消息，剩下近70%都是环境驱动的（通知25.8%、世界观测24.1%、静默突变19.9%）。

借鉴优化

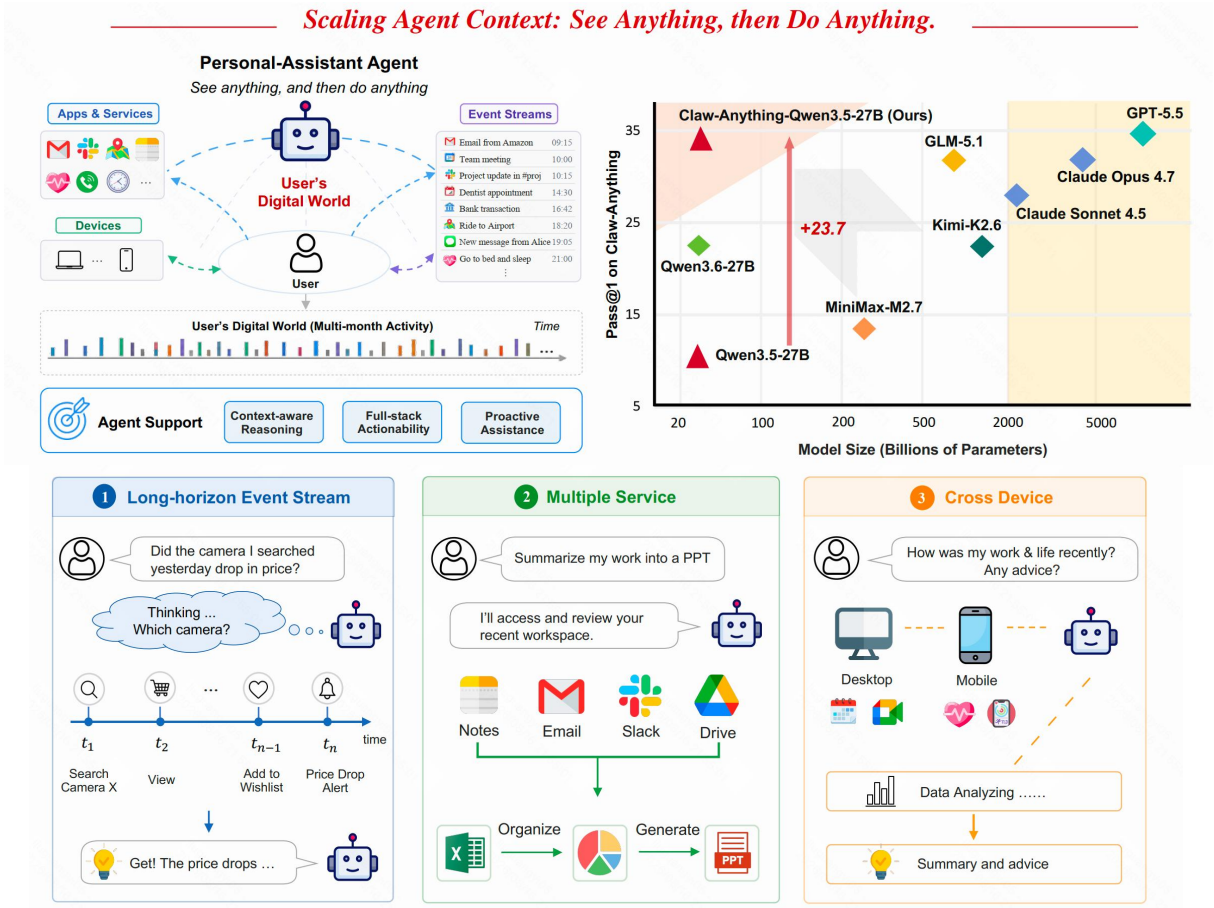
- ✓ 缺点
  - 所有服务都是模拟的，不是真实API，和真实世界的开放性有差距；
  - 任务是脚本化的固定时间线，真实生活的不确定性更强
- ✓ 主动性、动态世界适应、长程一致性的定义可借鉴
  - 整个任务从头到尾是同一个连续的多轮对话会话，所有触发回合的事件都是追加到现有上下文中，不会重置会话、不会新开对话。
  - 看下Gym是否是可复用的

Cowork Bench | Claw-Anything: Benchmarking Always-On Personal Assistants with Broader Access to User's Digital World (2606)

Yusong Lin[1,2,†] Xinyuan Liang[2,3,†] Haiyang Wang[2,✉] Qipeng Gu[2] Siqi Cheng[2]
Jiangui Chen[2] Shuzhe Wu[2] Feiyang Pan[2] Lue Fan[4] Sanyuan Zhao[1,✉] Dandan Tu[2,✉]

[1]Beijing Institute of Technology [2]Huawei Technologies Co., Ltd
[3]Peking University [4]Institute of Automation, Chinese Acade
✉Corresponding authors †Intern at Huawei

Fork 4 Star 47



贡献总结

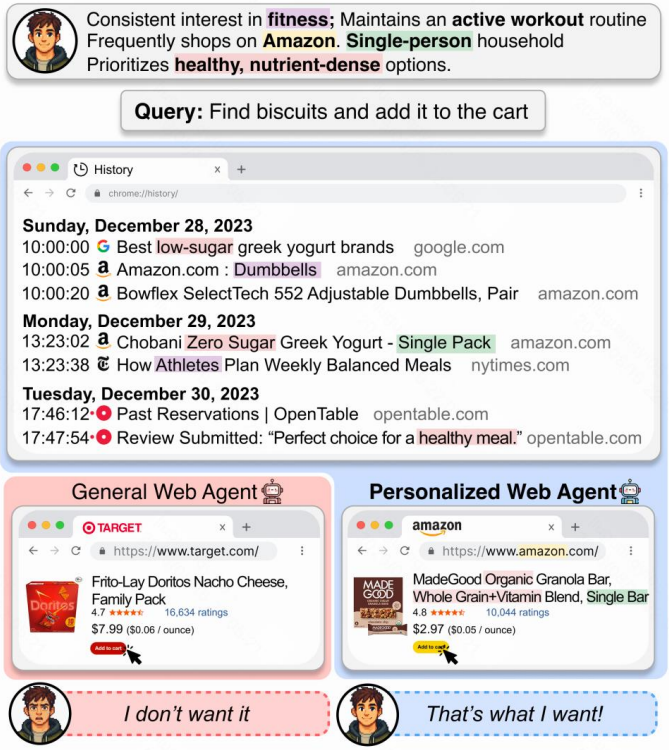
- ✓ 将个人助理Agent的评测从「单次孤立任务求解」推向「全场景长时在线辅助」，从时间、服务、设备三个维度扩展Agent的上下文边界，并配套了可规模化的自动化数据生成流水线
- ✓ Claw评测构建了一个mock环境模拟全场景长时在线辅助

借鉴优化

- ✓ 不涉及agent和human的模拟交互
- ✓ 主动服务是否有必要？
- ✓ 这里是预设好的环境，但是实际两者都得考虑
- ✓ 其实真实的情况下的上下文有两个方面，一方面是agent做了什么事情？另外一方面其实人在中间也会接管到电脑产生一些上下文
- ✓ 参考训练数据构建

Benchmark	Event Stream	Device Interfaces	# Services (avg. / max.)	Proactive	# Context Length by words	# Ins (Eval)	# Ins (Train)
ClawBench [31]	✗	CLI	1.6 (5)	✗	2.2k	313	0
WildClawBench [4]	✗	CLI	0.5 (3)	✗	2.6k	60	0
PinchBench [11]	✗	CLI	0.1 (3)	✗	1.7k	53	0
ClawMark [5]	✗	CLI	3.9 (5)	✗	2.0k	100	0
QwenClawBench [23]	✗	CLI	0.3 (6)	✗	12.1k	100	0
Claw-Eval [29]	✗	CLI	1.3 (6)	✗	5.3k	300	0
Claw-Anything (ours)	✓	**CLI + GUI**	**10.1 (18)**	✓	**191.7k**	**200**	**2000**

Web Agent Bench | Persona2Web: Benchmarking Personalized Web Agents in the Wild (2602)



[1]Department of Artificial Intelligence, Yonsei University, Seoul, Republic of Korea. Correspondence to: Dongha Lee <donalee@yonsei.ac.kr>.

Proceedings of the 43rd International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026. Copyright 2026 by the author(s).

Fork 1 Star 3

Benchmark	Domains	Environment	Personalization
WebLINX	–	Cached (155)	✗
Mind2Web	31	Cached (137)	✗
WebShop	1	Simulated	✗
WebArena	6	Simulated	✗
WebVoyager	5	Open web (15)	✗
AssistantBench	15+	Open web (258)	✗
WebCanvas	19	Open web (69)	✗
PersonalWAB	1	Function-call	✓
Apollonion	6	Dialogue agent	✓
PERSONA2WEB	**21**	**Open web (105)**	✓

贡献总结

- ✓ 面向真实开放网络环境、专门评估 Web Agent 个性化推理能力的基准
- ✓ Web agent操作领域通过预设个性化的操作历史，评测个性化任务的执行的能力
- ✓ 知识库是预设的，类似lomoco的知识库的形式【纯生成的文本描述形式的操作】
- ✓ 总量：50 个用户，共 102,568 条历史，人均 2051 条，人均覆盖 35 个网站。

借鉴优化

- ✓ – 借鉴
  - 任务失败了，到底是导航能力差，还是个性化能力差？需要设计更加细粒度的指标
  - 文本生成、对话场景（如 LongMemEval），核心是基于用户资料生成个性化回复
  - 如果我们也要采取预设操作历史的形式，那么文中的 按需访问（On-demand） 和 执行前访问（Pre-execution） 实验方案就可以借鉴
- ✓ 优化
  - 个性化的瓶颈不止是「检索不到历史」，「检索到了但不会用」同样是重大障碍。 => 模型训练目标。
  - 不涉及经验知识、不涉及多轮交互

Web Agent Bench | LongMemEval-V2: Evaluating Long-Term Agent Memory Toward Experienced Colleagues (2605)

Di Wu* **Zixiang Ji** **Asmi Kawatkar** **Bryan Kwan**
Jia-Chen Gu **Nanyun Peng** **Kai-Wei Chang**
University of California, Los Angeles

https://xiaowu0162.github.io/longmemeval-v2/

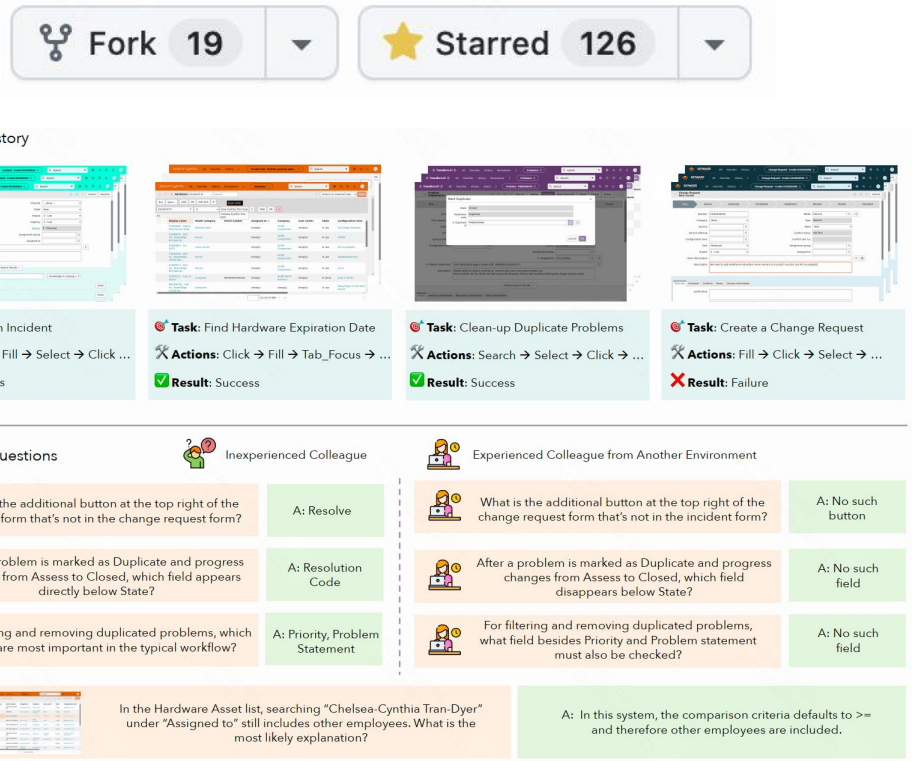


Figure 1: Examples of LongMemEval-V2 questions. We display examples from the WorkArena domain. LME-V2 questions exercise diverse memory abilities: static (row 1), dynamic (row 2), workflow (row 3), gotchas (row 4), and premise awareness (right column).

贡献总结

- ✓ 是LongMemEval系列的V2版本。相比V1聚焦对话式个人记忆，V2将评测对象转向Web智能体的环境经验记忆，核心命题是：高质量的长期记忆，应当让Agent成为特定定制环境中的「有经验的同事」
- ✓ web场景收集了很多离线的成功和失败的轨迹，标注了400个QA任务

借鉴优化

- ✓ 借鉴
  - 环境记忆这个词比较好，提出一些新的学术概念
  - 部分Agent记忆工作（agent-s2）通过下游任务成功率间接衡量记忆效果，无法区分是记忆差、规划差还是执行差，难以定位问题。
  - 记忆模块本身就是Agent的「外置记忆」，只负责提供证据，不负责最终决策
  - 那完全可以把这套逻辑迁移到ubuntu上了，其实最终还是问答式的，直接ubuntu上收集个性化的轨迹 + 成功和失败轨迹都成 QA任务 + 可再补充执行类的任务优化
- ✓ 优化
  - 主要是去验证验证「跨多条轨迹积累环境经验」的能力，不涉及到用户画像，也不涉及具体的任务执行

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① 实习的核心原则 + 前边说的两个方向的选择

② 为啥Lifelong CUA是重要的并且可行的？

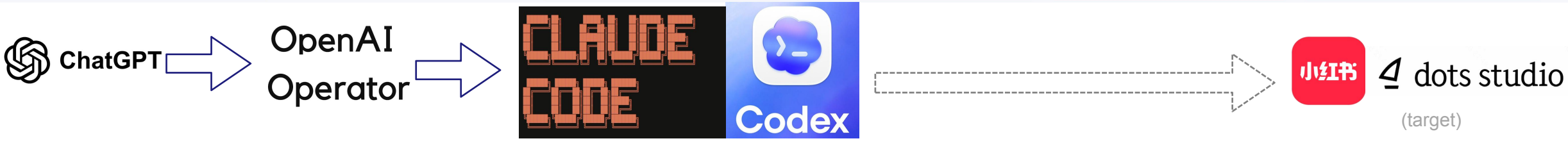
③ Lifelong CUA这个领域相关的研究工作有哪些？各有哪些痛点问题？有哪些可借鉴的点？

④ 当前的CUA距离最终的LifelongCUA还有哪些差距？分阶段给出研究目标和研究计划

⑤ 为什么我能够胜任这份工作

Lifelong CUA Maturity Model

CAPABILITY VECTOR A / C / E / P / W / I / R Context → memory → service Cell = coverage status + shortest defining behavior	L0 Isolated Task 独立任务执行 task ends · state resets	L1 Stateful Episode 有状态单次任务 episode-only state	L2 Contextual Persona 上下文式个性化 static / preloaded context	L3 Persistent CUA 持久纵向 CUA cross-session persistence	L4 Lifelong Personal CUA 终身个性化 CUA online cross-task learning
	0 / 14	6 / 14	8 / 14	11 / 14	14 / 14
CONTEXT SOURCES A Agent-mediated History	No history — NONE	Episode trace — PARTIAL	Preloaded logs — PARTIAL	Persistent history — EXPLICIT	Online task history — EXPLICIT
C Computer / World Changes	Current state only — NONE	In-episode changes — PARTIAL	Fixed snapshot — PARTIAL	Periodic change log — PARTIAL	Continuous observation — EXPLICIT
MEMORY & PERSONALIZATION E Experience Memory	No reuse — NONE	Working memory — PARTIAL	Offline examples — PARTIAL	Persistent episodes — EXPLICIT	Online experience learning — EXPLICIT
P User Profile Memory	No profile — NONE	No persistent profile — NONE	Static persona — EXPLICIT	Persistent profile — EXPLICIT	Dynamic user profile — EXPLICIT
W World-State Memory	No persistent state — NONE	Episode state — PARTIAL	Fixed personal world — PARTIAL	Persistent snapshots — PARTIAL	Fresh world-state model — EXPLICIT
CONTINUOUS SERVICE I Continuous Interaction	One-shot — NONE	Multi-turn episode — EXPLICIT	Context-aware — PARTIAL	Cross-session resume — EXPLICIT	Continuous collaboration — EXPLICIT
R Proactive Service	Reactive only — NONE	Reactive only — NONE	Contextual suggestion — PARTIAL	Limited triggers — PARTIAL	Event + goal driven — EXPLICIT
MEMORY LIFECYCLE write · retrieve · update · consolidate · correct · forget	Reset after task	Session read / write discarded at episode end	Retrieval-oriented mostly static context	Persistent + periodic consolidate / correct	Online full lifecycle learn · update · correct · forget



Benchmark & Harness | LifelongCUA-Bench-v1

论文题目

- LifelongCUA-Bench: Benchmarking Experience Memory and Personalized Interaction in Computer-Use Agents

主要要做的事情

- 在不去考虑复杂动态环境的情况下以预设知识库的形式去评测目前的模型/harness对于经验记忆和用户的个性化操作历史的理解和使用能力

主要解决的已有工作的痛点问题

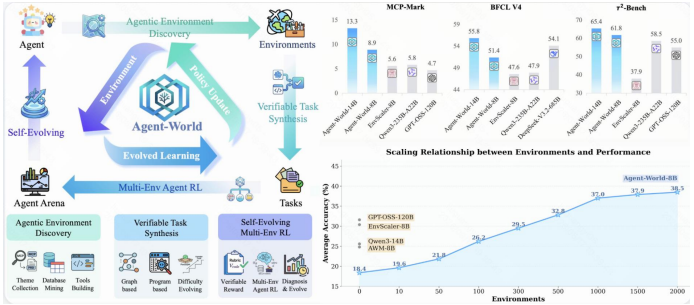
- osworld/ALE 仅评测单次任务执行，不涉及经验记忆和用户的个性化的环境下的记忆管理与任务执行

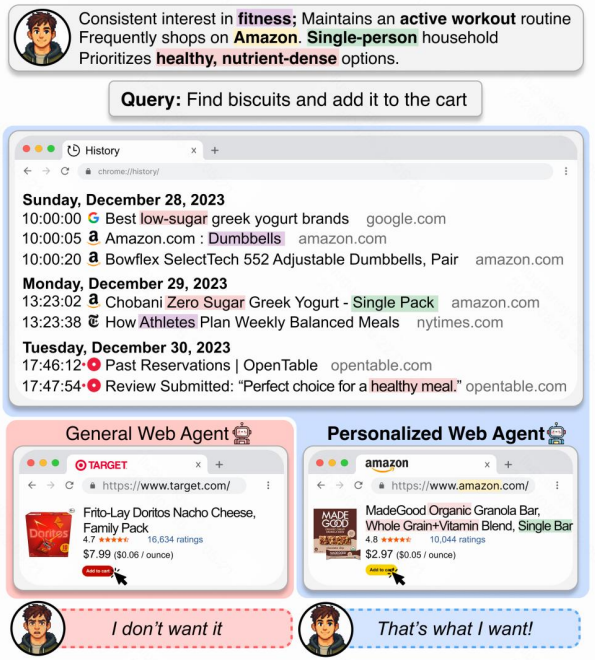
论文主要的contribution

- 在computer use场景下构造经验记忆库和个性化操作历史，全面评测agent的在CUA场景下的记忆能力
- 两种评测方式：给定经验库进行QA + 动态任务执行
- 两种评测内容：经验记忆评测 + 个性化环境与操作历史的评测
- 评测haeness：codex + openclaw + 自建

可能会遇到的卡点/困难

- 构造真实/有意义的用户数据（环境）





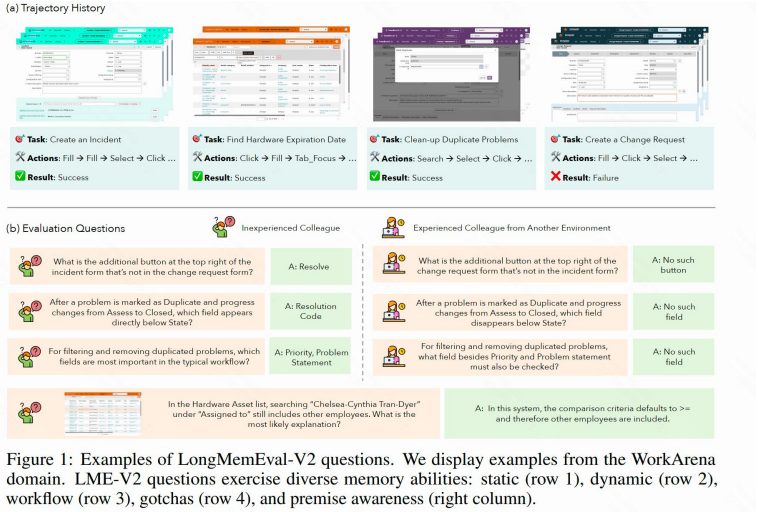


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Benchmark & Harness | LifelongCUA-Bench-v2

论文题目

- **LifelongCUA-Bench-v2: Benchmarking Longitudinal Human–Agent Collaboration in Evolving Computer Environments**

主要要做的事情

- 构造稳定的动态环境与人类用户，模拟人和CUA的multi—tern、multi—session的设定下的交互

主要解决的已有工作的痛点问题

- 经验记忆和用户偏好记忆都是静态的，同时不包含computer环境的动态变化以及human与CUA的cowork

论文主要的contribution

- 在computer use场景下评测human和CUA的multi—tern、multi—session设定下的cowork能力
- 评测haeness：codex + openclaw + 自建

可能会遇到的卡点/困难

- 构造真实/有意义的用户数据（环境）
- 模拟human和CUA的cowork的实现
- 可同时参考ClawMark和VibeLifeBench引入世界的变化

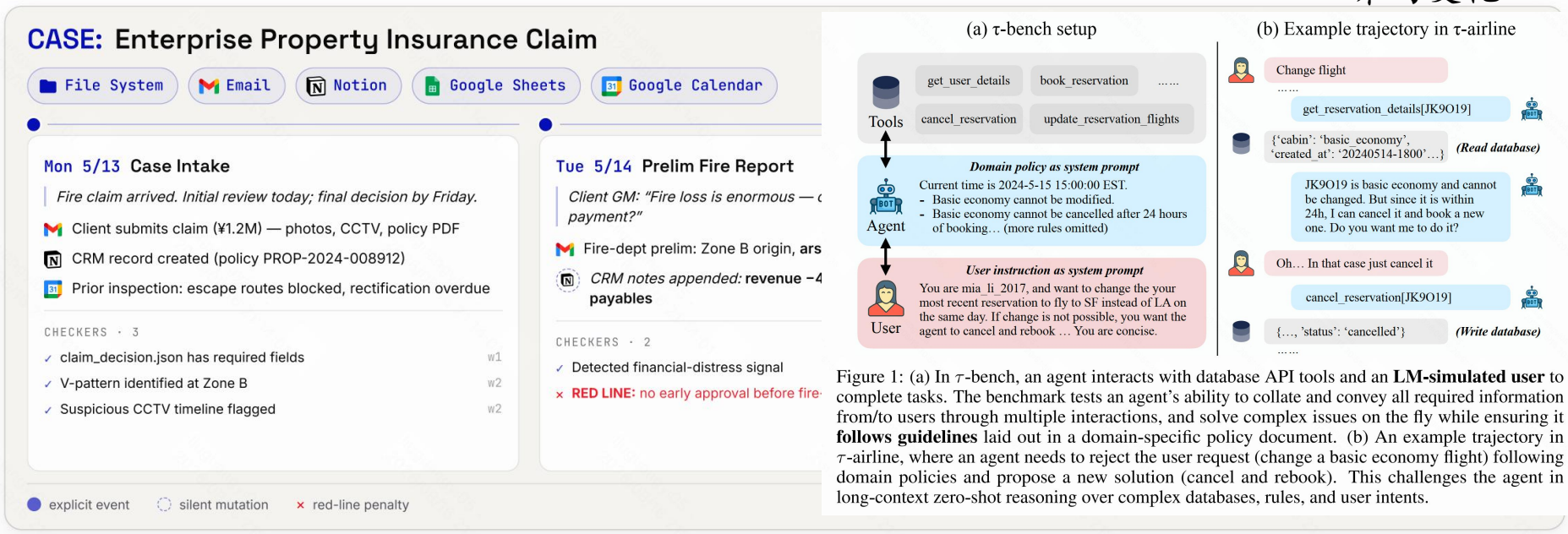
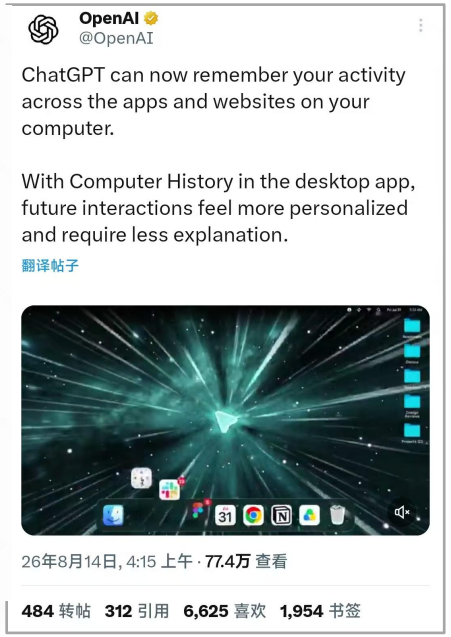


Figure 1: (a) In τ-bench, an agent interacts with database API tools and an **LM-simulated user** to complete tasks. The benchmark tests an agent's ability to collate and convey all required information from/to users through multiple interactions, and solve complex issues on the fly while ensuring it **follows guidelines** laid out in a domain-specific policy document. (b) An example trajectory in τ-airline, where an agent needs to reject the user request (change a basic economy flight) following domain policies and propose a new solution (cancel and rebook). This challenges the agent in long-context zero-shot reasoning over complex databases, rules, and user intents.



Datasets and Model

0 - 3 个月

LifelongCUA-Bench V1

问题定义
验证静态经验知识
验证用户偏好知识

3 - 6 个月

LifelongCUA-Bench V2

搭建动态环境
模拟人和CUA的持续交互
确定harnesss

6 - 9 个月

Personal Context Dataset

数据构建管线
专家轨迹采集
合成数据飞轮

9 - 12 个月

Lifelong CUA Model

训练与策略学习
真实应用扩展
技术报告

Benchmarks	DeepSeek-V4-Pro-0813	DeepSeek-V4-Flash-0731	DeepSeek-V4-Pro-Preview	DeepSeek-V4-Flash-Preview	GLM-5.2	Kimi-K3	Opus-4.8	Fable 5 (w/ fallback)
HLE (wo/w tools)	42.7/60.0	37.8/51.5	37.7/48.2	34.8/45.1	40.5/54.7	43.5/56.0	49.8/57.9	53.3/63.0
Terminal Bench 2.1	87.9	82.7	72.1	61.8	81.0	88.3	85.0	88.0
NL2Repo	61.5	54.2	38.5	39.4	48.9	-	69.7	-
Cybergym	83.3	76.7	52.7	38.7	-	80.0	78.3	83.1
DeepSWE	62.7	54.4	12.8	7.3	46.2	67.5	58.0	70.0
Toolathlon-Verified	74.1	70.3	55.9	49.7	59.9	76.5	76.2	77.9
Agents' Last Exam	25.7	25.2	16.5	15.8	23.8	27.6	25.7	-
AutomationBench (Public)	31.8	25.1	12.8	10.8	12.9	30.8	27.2	29.1
DSBench-FullStack	71.1	68.7	41.8	37.0	61.8	73.7	71.6	77.2
DSBench-Hard	67.2	59.6	31.1	25.8	54.5	63.0	71.7	68.3

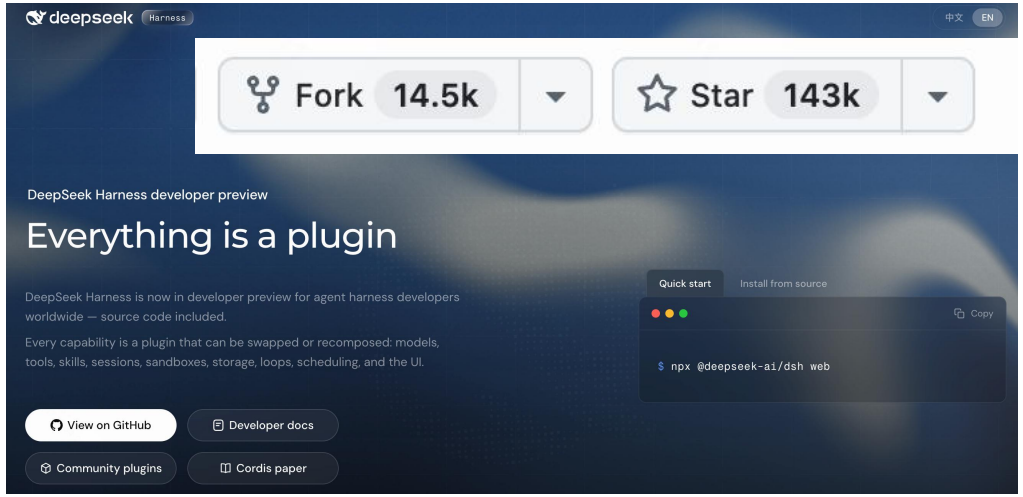


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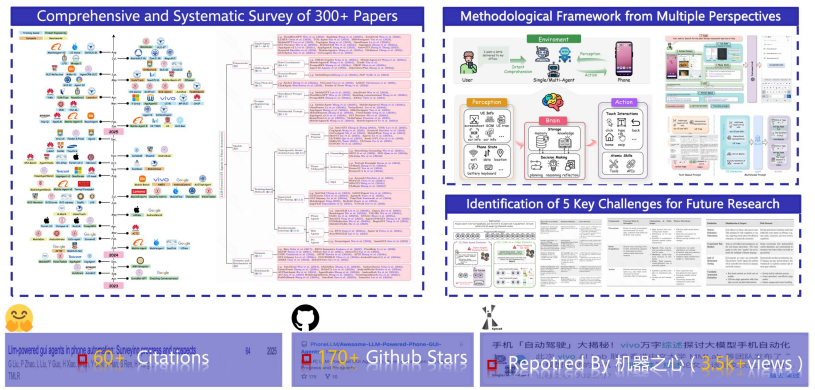
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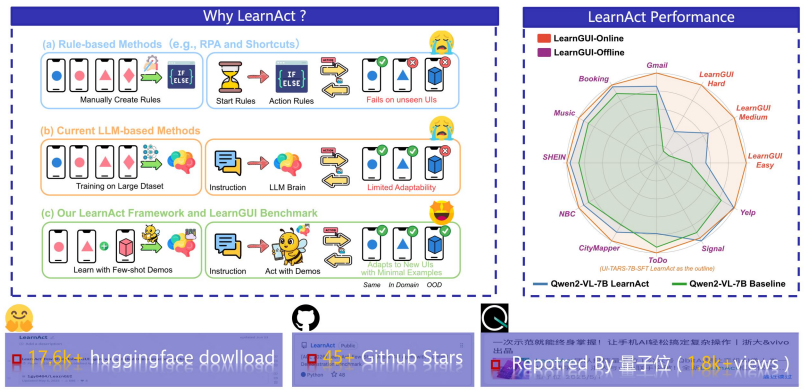
为什么我能胜任这份工作?

[TMLR] LLM-Powered GUI Agents in Phone Automation: Surveying Progress and Prospects



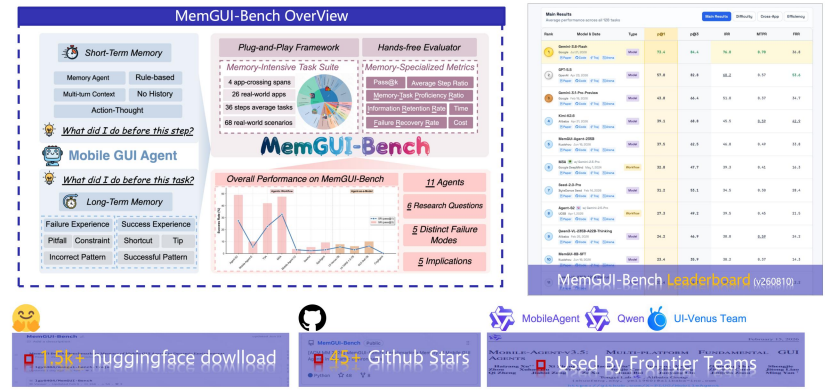
领域认知、深入用户调研

[ACL 2026] LearnAct: Few-Shot Mobile GUI Agent with a Unified Demonstration Benchmark



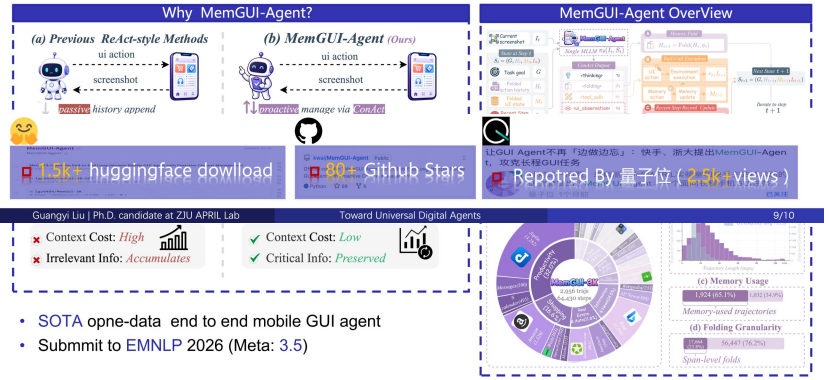
benchmark构建、数据采集、后训练

[ACMMM 2026] MemGUI-Bench: Benchmarking Memory of Mobile GUI Agents in Dynamic Environments



benchmark构建、数据采集、memory

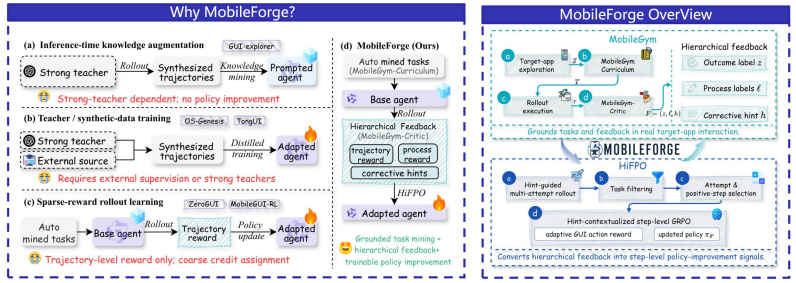
[Arxiv 2026] MemGUI-Agent: An End-to-End Long-Horizon Mobile GUI Agent with Proactive Context Management



- SOTA opne-data end to end mobile GUI agent
- Submmit to EMNLP 2026 (Meta: 3.5)

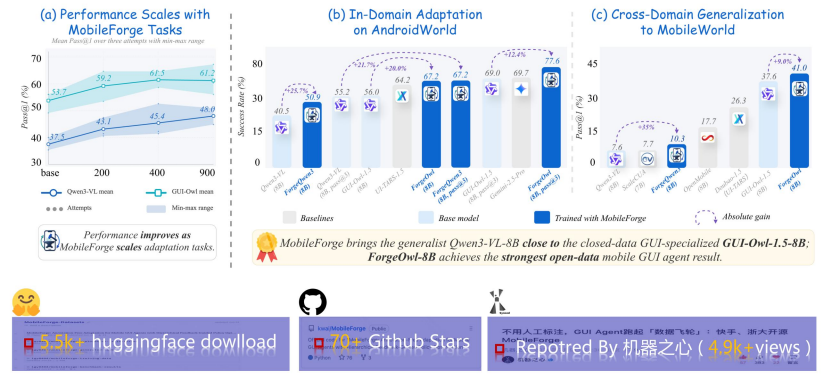
harness、数据采集、后训练、memory

[Arxiv 2026] MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization



- MobileForge turns real target-app interaction into executable curricula, hierarchical rollout feedback, and hint-contextualized policy updates without human-written tasks, demonstrations, or reward labels.

[Arxiv 2026] MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization



数据采集、大规模rollout环境搭建、Agentic RL、后训练

其他向国海老师请教的点？

01	**V1 边界**	先做一个极长程工作流，还是从第一天就覆盖跨 Session 连续性？
02	**执行环境**	先做 Web / Mock，还是直接改造 OSWorld / CUA-Gym 风格的 Ubuntu 环境？
03	**资源边界**	第一季度可获得多少工程、Rollout 与专家数据支持？
04	**成功标准**	3 个月交付论文级 Benchmark + Harness；3–6 个月形成模型 / 策略结果？

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